VOLTAGE HEADROOM / USABLE FLIGHT TIME

Chiralight standard-motor

Fly longer. Carry more.

More of your battery is useful when the motor drive retains thrust at lower voltage. This simulated 3 kg quad uses 10-inch props, a 640KV motor and a 6S 4 Ah pack, while retaining a **3:1 thrust-to-weight ratio (TWR)** maneuver reserve for gust recovery and obstacle avoidance.

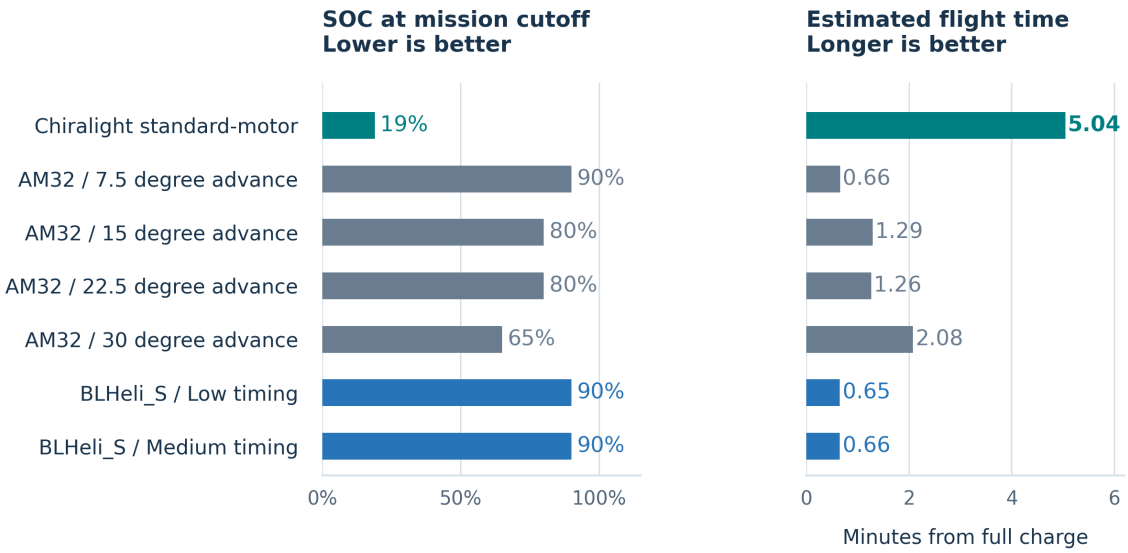
5.0 min
Estimated flight time
Chiralight standard-motor

+143%
Flight time versus the
best tested alternative

+174 g
Potential payload by
choosing 6S over 8S

3:1 TWR MANEUVER RESERVE FOR GUST RECOVERY AND OBSTACLE AVOIDANCE

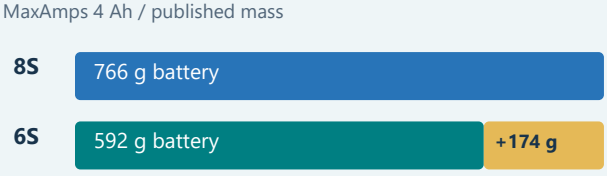
640KV / 6S | 1.5:1 TWR flight load | 3:1 TWR maneuver reserve



AM32 and BLHeli_S show every tested timing setting. The headline compares against **AM32 / 30 degree advance**, the longest-flying tested alternative in this configuration.

6S FOR PAYLOAD. 8S FOR ADDITIONAL VOLTAGE RESERVE.

The smaller 4 Ah pack releases **174 g for payload** at the same 3 kg all-up mass. With Chiralight standard-motor, 8S estimates 7.91 minutes and adds 174 g.



Matching energy with a lower-Ah 8S pack can leave insufficient power for the 3:1 TWR reserve.

Flight-time rule: battery discharge uses the measured power for a 1.5:1 TWR equivalent flight load. Discharge stops at the lowest tested SOC that meets both required maneuver reserve and 3.0 V/cell under load. The higher-thrust check includes its own battery sag; maneuver power is not charged continuously.

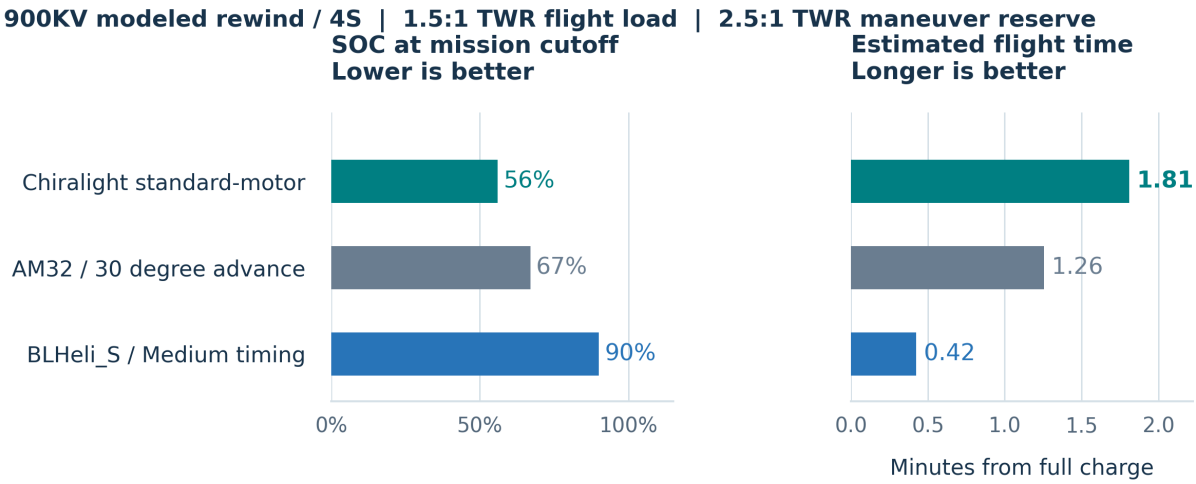
Verifiable simulation results. Contact Chiralight Systems for details.
Evaluate for your aircraft. Exact performance may vary.

CONFIGURATION TRADEOFFS / REPRODUCIBLE RESULTS

Match the drive to the mission.

Modeled 900KV rewind / 4S: lower battery mass

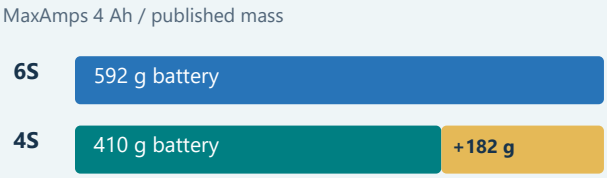
The lower-S, higher-KV option trades maneuver reserve for battery mass: 2.5:1 TWR, rather than 3:1 TWR. At the same 1.5:1 TWR equivalent flight load, Chiralight standard-motor gives **1.81 minutes** versus **1.26 minutes** for its best tested alternative.



Each competitor uses its best tested setting for flight time in this configuration. The 4S reserve load averages 145 A (164 A peak), above the pack's 128 A continuous rating.

4S FOR PAYLOAD. 6S FOR ADDITIONAL VOLTAGE RESERVE.

The smaller 4 Ah pack releases **182 g for payload** at the same 3 kg all-up mass. The larger pack stores more energy and gives the drive more voltage headroom.



FLIGHT POWER AND MANEUVER RESERVE ARE DIFFERENT REQUIREMENTS

1.5:1 TWR flight load: 1,125 gf per rotor supplies the mission-power estimate.
3:1 TWR maneuver reserve: 2,250 gf per rotor is the 640KV maneuver gate.
2.5:1 TWR reduced reserve: 1,875 gf per rotor is the explicitly reduced 900KV gate.
All four motors share the battery. Each reserve check includes the higher current and loaded-voltage sag.

Built from running code. The bench runs compiled Chiralight standard-motor, AM32 and official BLHeli_S code against the same nonlinear B/H motor plant and EMAX-table-informed propeller load. Flight-power points feed a battery-discharge replay; reserve cutoffs come from separately executed thrust tests. The model uses 4 Ah packs, 5 milliohms per series cell, a synthetic LiPo OCV/SOC curve, four motors and 10 W auxiliaries. FOC uses 1 A injection, MTPA and overmodulation with dedicated field weakening disabled.

Published battery references: MaxAmps LiPo 4 Ah 4S, 410 g / 6S, 592 g / 8S, 766 g. Contact Chiralight Systems for source traces, controller settings and reproduction instructions.